

README.md

PROJECT TITLE:

Password Strength Checker

OVERVIEW OF THE PROJECT:

The Password Strength Checker is a Python-based command-line application designed to evaluate the security strength of a user-provided password. It analyzes password length, uppercase and lowercase letters, digits, special characters, whitespace, common password patterns, keyboard sequences, common names, and repeated characters. The program calculates a security score out of 10, assigns a strength rating, and provides actionable feedback to improve the password.

FEATURES:

- Password Character Analysis: Counts uppercase letters, lowercase letters, digits, special characters, and whitespace.
- Length Checking: Checks whether the password meets minimum length requirements and gives additional score for longer passwords.
- Common Password Detection: Detects commonly used passwords or password patterns such as password, 123456, qwerty, and similar entries.
- Keyboard Pattern Detection: Identifies sequential keyboard patterns such as qwerty, asdfgh, zxcvbn, 123456, and 654321.
- Common Name Detection: Checks whether easily guessable names are included in the password.
- Repeated Character Detection: Detects three or more identical characters appearing consecutively.
- Security Score: Calculates and displays a score from 0 to 10.
- Strength Rating: Classifies passwords as Very Weak, Weak, Moderate, Strong, or Very Strong.
- Visual Score Meter: Displays a simple # and - based visual representation of the security score.
- Actionable Feedback: Provides suggestions and warnings based on detected weaknesses.
- Report Saving: Allows the user to save the evaluation report to a text file named saved_passwords.txt.

TECHNOLOGIES/TOOLS USED:

- Programming Language: Python 3
- Libraries/Modules: os (imported in the program)
- Core Concepts: Strings, Lists, Loops, Conditional Statements, Boolean Flags, Counters, User Input, File Handling
- Environment: Any terminal or IDE supporting Python, such as VS Code, PyCharm, or Command Prompt

STEPS TO INSTALL & RUN THE PROJECT:

- 1 Ensure Python 3.x is installed on your computer.
- 2 Save the Python source code as main.py in your working directory.
- 3 Open the terminal or command prompt and navigate to the project directory:

```
cd path/to/your/folder
```

Run the program using:

```
python main.py
```

INSTRUCTIONS FOR TESTING:

- 1 **Test a Strong Password:** Enter a password containing at least 12–16 characters, uppercase and lowercase letters, digits, and special characters. Verify the score and strength rating.
- 2 **Test a Short Password:** Enter a password with fewer than 8 characters and confirm that the program reports that it is too short.
- 3 **Test Missing Character Types:** Enter passwords without uppercase letters, lowercase letters, digits, or special characters and verify that appropriate suggestions are displayed.
- 4 **Test Common Passwords:** Enter a password containing a known common pattern such as 'password' or '123456' and verify that a warning and score reduction are applied.
- 5 **Test Keyboard Patterns:** Use a password containing patterns such as 'qwerty' or 'asdfgh' and verify the keyboard-pattern warning.
- 6 **Test Common Names:** Enter a password containing one of the checked common names and verify the corresponding warning.
- 7 **Test Repeated Characters:** Enter a password containing three or more identical characters consecutively and verify that the repeat warning appears.
- 8 **Test Report Saving:** Choose 'yes' when asked to save the report and confirm that saved_passwords.txt is created or updated.

OUTPUT / EVALUATION REPORT:

After evaluating the password, the program displays the password length, character counts, security score, visual score meter, strength rating, and any actionable suggestions or warnings.

REPORT STORAGE:

If the user chooses to save the report, the program appends an audit entry to saved_passwords.txt.

The saved entry contains the entered name, registration number, evaluated password, password statistics, security score, strength rating, visual meter, and feedback.

SCREENSHOTS:

```
=====
                        PASSWORD STRENGTH CHECKER
=====
Enter Your Name: Ansh Walia
Enter Your Registration Number: 26BAI10538
Enter a password to evaluate: HeyIt$J10

=====
                        EVALUATION REPORT
=====
Password Length      : 9
Uppercase Count      : 3
Lowercase Count      : 3
Digit Count          : 2
Special Char Count    : 1
Whitespace Count      : 0

-----
Security Score        : 6 / 10
Score Visualizer      : [#####----]
Strength Rating       : Moderate
-----
Excellent! Your password passed all security checks.

=====
Do you want to save this report to a file? (yes/no): yes
[+] Report successfully saved to saved_passwords.txt
PS C:\Users\Ansh walia\OneDrive\Documents\Codes> |
```

```
=====
                        PASSWORD STRENGTH CHECKER
=====
Enter Your Name: Billu Don
Enter Your Registration Number: 25
Enter a password to evaluate: Billu_Ke_Changu1_se_n@_bach_paoqe

=====
                        EVALUATION REPORT
=====
Password Length      : 33
Uppercase Count      : 3
Lowercase Count      : 22
Digit Count          : 1
Special Char Count    : 7
Whitespace Count      : 0

-----
Security Score        : 7 / 10
Score Visualizer      : [#####---]
Strength Rating       : Strong
-----
Excellent! Your password passed all security checks.

=====
Do you want to save this report to a file? (yes/no): yes
[+] Report successfully saved to saved_passwords.txt
PS C:\Users\Ansh walia\OneDrive\Documents\Codes> |
```